PRECISION ACCOUNTING FOR KV AND RECURRENT-STATE CACHES IN HYBRID LLM SERVING

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ABSTRACT

Hybrid linear-attention models allocate two caches with different scaling laws. Attention key-value (KV) storage grows with context, while recurrent state is fixed per sequence. We characterize their joint precision budget in Qwen3.5-2B/9B with vLLM on one RTX 5090. This is an allocator study, not a claim about scheduler admission or end-to-end benefit. Across a frozen 112-cell matrix, all 52 paired fp32/bf16 state comparisons increase allocator token capacity. The median increase is 15.44%. A byte-derived continuous model has 2.38% median absolute residual, with signed residuals from -5.16% to $+13.21\%$. Discrete block and recurrent-page accounting explains why the model is neither a lower nor an upper bound. For 2B, int4 KV, and 4K context, the reported capacities correspond to 657.4 and 904.3 allocator-equivalent sequence slots. These are not demonstrated concurrent requests. We reanalyze GSM8K as 1,800 paired seed-item draws with two-way clustering by item and dataset seed. State-only changes accuracy by -1.00 percentage point (95% CI $[-2.04, +0.04]$), while int4 KV changes it by -2.72 points ($[-5.28, -0.17]$). Under these corrected intervals, an offline selector chooses full precision for the strict request; the medium, high, and negative-control requests have no feasible candidate. A broader serving matrix also fails its frozen continuous-goodput stability gate. The supported contribution is therefore byte-exact allocator characterization with an auditable, fail-closed decision trace.

1 INTRODUCTION

KV-cache quantization is a standard way to reduce the memory cost of large language model serving (Liu et al., 2024; Hooper et al., 2024; Sharma et al., 2024). Most analyses assume an attention-only transformer. In that setting, every cache layer grows with the number of retained tokens.

Hybrid linear-attention models violate this assumption. Qwen3.5 interleaves Gated Delta Network (GDN) layers with grouped-query attention (GQA) layers (Yang et al., 2025; Team, 2026). Each sequence carries a fixed recurrent state in addition to its attention KV. Both allocations draw from one vLLM cache pool (Kwon et al., 2023). KV precision changes the per-token term, while state precision changes the per-sequence term.

The state data type is already a vLLM deployment switch (`mamba_ssm_cache_dtype`) (vLLM Project, 2025). We do not introduce a new quantizer, cache allocator, or online scheduler. Our contribution is a scoped empirical characterization of how existing KV and state controls compose. We make four contributions:

1. We derive byte-exact attention and recurrent-state terms, then connect them to vLLM's discrete hybrid-cache allocation rule.

2. We validate this accounting on two temporal 112-cell allocator runs and report residuals without converting them into a bound.

3. We replace seed-iid GSM8K inference with paired seed-item analysis that clusters by both item and dataset seed.

4. We replay a four-candidate selector with complete constraint traces. The corrected evidence leaves only the strict request feasible, at full precision.

The target regime is short-context, memory-limited deployment, where fixed state can consume a substantial fraction of the cache pool. Our conclusions are limited to Qwen3.5, vLLM, one RTX 5090, and the measured strata. We do not infer usable concurrency or stable serving gains from allocator capacity.

2 BACKGROUND AND POSITIONING

KV methods reduce precision or retain fewer tokens (Liu et al., 2024; Hooper et al., 2024; Sharma et al., 2024; Zhang et al., 2023; Xiao et al., 2024; Li et al., 2024). State-space

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Table 1. **Capability boundary.** “Existing” means this work uses an available mechanism rather than claiming it as new.

Approach	KV	State	Audit
KV quant. (Liu et al., 2024; Hooper et al., 2024)	yes	no	no
State compression (Dao, 2026)	no	yes	no
vLLM controls (Kwon et al., 2023; vLLM Project, 2025)	yes	yes	no
This work	existing	existing	yes

and hybrid models instead maintain recurrent state (Gu & Dao, 2024; Dao & Gu, 2024; Lieber et al., 2024; Botev et al., 2024; Glorioso et al., 2024). State compression and lower-precision state paths are also prior art (Dao, 2026; vLLM Project, 2025; 2026). These mechanisms are complementary. One changes A, the per-token attention cost, while the other changes G, the fixed state cost.

Table 1 states the novelty boundary directly. The new artifact is the byte-exact joint accounting, complete allocator matrix, dependence-aware quality audit, and fail-closed decision trace. It is not the underlying precision switches. This positioning is narrower than a new serving system, but it is testable from the supplied evidence.

3 Allocator and Decision Models

Let L be context length, A the per-token attention-KV bytes, G the fixed recurrent-state bytes per sequence, and M the available cache bytes. The continuous approximation is

$$\widetilde{S}(L) = \frac{M}{AL+G}, \qquad \widetilde{T}(L) = L\widetilde{S}(L). \tag{1}$$

Here $\widetilde{S}$ is an idealized sequence-slot count. It is not scheduler admission. The predicted state ratio is

$$r_{\text{state}}(L) = \frac{AL+G_{\text{fp32}}}{AL+G_{\text{bf16}}}. \tag{2}$$

The ratio decreases as L grows because the attention term becomes dominant.

The engine reports a discrete capacity. Let K be the allocated GPU-block count, H the number of recurrent cache groups, and B the attention block size in tokens. For this hybrid layout,

$$\begin{aligned} S_{\text{alloc}}(L) &= \frac{K}{H+\lceil L/B \rceil}, \\ T_{\text{alloc}}(L) &= \lfloor L S_{\text{alloc}}(L) \rfloor . \end{aligned} \tag{3}$$

This equation reconstructs the reported token capacities. The Supplementary Material provides the full parameter and padding derivation.

For deployment stratum x and precision candidate q, the offline selector uses

$$q^*(x) = \arg \underline{\max_{q\in\mathcal{F}(x)} \text{Goodput}}(q,x). \tag{4}$$

The feasible set requires exact profile matches. It checks a quality lower bound, latency upper bounds, cache bytes, and $S_{\text{alloc}} \geq S_{\text{min}}$. Missing or ambiguous rows fail closed. The selector never labels S_{alloc} as operational concurrency.

4 Evaluation

Platform and allocator runs. We evaluate Qwen3.5-2B and 9B on one RTX 5090 with 32 GB memory. The frozen vLLM worktrees are identified by commits `55f4768` and `e2fa285`. The formal allocator grid crosses model, KV data type, state data type, context length, and memory utilization. Two temporal runs each complete 112 cells, including 52 fp32/bf16 core pairs. Allocator cells are deterministic for a frozen build and configuration; the second attempt checks temporal reproducibility rather than statistical replication.

Quality and statistical units. The GSM8K protocol uses greedy no-think decoding and samples 200 items without replacement for each of nine dataset seeds (Cobbe et al., 2021). The same seed selects the same ordered items for every precision candidate. The 1,800 seed-item draws contain 1,017 unique items; pairwise seed overlap ranges from 19 to 39. We fit an intercept-only paired contrast with two-way CR1 clustering by item and dataset seed. Confidence intervals and nominal two-sided p values use a Student-t reference with 8 degrees of freedom. A 20,000-replicate bootstrap over unique items provides an equal-item-weight sensitivity analysis. The nine seed summaries are descriptive, not iid replicates. This CPU reanalysis uses Python 3.13.5, NumPy 2.1.3, SciPy 1.15.3, and statsmodels 0.14.4.

Perplexity uses a chunk-level C4/PG19 harness with three dataset seeds. RULER uses five preregistered model-task-length cells, 20 samples per cell, and three dataset seeds (Hsieh et al., 2024). Exact zero RULER differences are reported as observed equality, not equivalence.

Selector and serving evidence. The selector profile contains four candidates: full, KV-only, state-only, and joint. Each candidate stores physical cache bytes, allocator-equivalent slots, serving bounds, and dependence-aware GSM8K intervals. We replay four frozen requests without changing their thresholds after observing the reanalysis.

Serving uses Random60 and ShareGPT300 workloads (LMSYS, 2023), three seed and trace repeats, Poisson arrivals, and fixed measurement windows. A request is good only if it completes and satisfies the specified time-to-first-token (TTFT) and time-per-output-token (TPOT) limits. SLO-goodput is the number of good requests divided by measurement duration. Failed requests remain in the denominator. A second run uses the same host, contracts, and seeds, so we call it a temporal rerun.

Table 2. **Fixed-utilization allocator slice.** Token columns are total allocator token capacity. Gap is $(r_{\text{measured}} - r_{\text{model}})/r_{\text{model}}$. The 9B/fp16/16K cell was not probed in the frozen 0.85 fp16 attempt; it is not silently imputed.

KV	Cell	fp32 total	bf16 total	ratio	gap
int4	2B/4K	2,692,710	3,703,954	1.3755	-2.37%
int4	2B/16K	4,895,837	5,458,458	1.1149	-3.24%
int4	9B/4K	315,392	443,538	1.4063	-0.19%
int4	9B/16K	573,440	653,635	1.1398	-1.07%
fp16	2B/4K	1,199,383	1,384,448	1.1543	-0.16%
fp16	2B/16K	1,552,143	1,661,337	1.0704	$+2.46\%$
fp16	9B/4K	144,104	161,899	1.1235	-2.83%

5 Results

5.1 Allocator capacity

Across the formal 52-pair matrix, every bf16-state cell has higher allocator token capacity than its fp32-state pair. The median gain is 15.44%, and the range is 3.32–93.28%. The largest gains occur in short-context cells where fixed state dominates. The temporal rerun reproduces all 112 cells within the frozen token tolerance; the largest per-cell difference is 1.42%.

The corrected continuous model has 2.3795% median absolute residual. Signed residuals range from -5.1604% to $+13.2134\%$. Their mixed signs rule out a lower-bound or upper-bound interpretation. Figure 1 compares the model with the fixed-utilization slice.

The byte correction matters. Each GDN layer contains a 36,864-byte bf16 convolution state plus a temporal state. Changing only the temporal state from fp32 to bf16 reduces per-layer bytes from 1,085,440 to 561,152, or 48.30%. It does not halve the complete GDN state. Figure 2 shows how the smaller recurrent page changes discrete block sizes and counts.

For 2B/int4/4K, fp32 uses $K = 3287$, $B = 2064$, and $H = 3$. Equation (3) gives $3287/(3+2) = 657.4$ slots and 2,692,710 tokens. Bf16 uses $K = 6330$ and $B = 1072$, giving $6330/(3+4) = 904.3$ slots and 3,703,954 tokens. The difference is 246.9 allocator-equivalent slots. We have not shown that a scheduler admits or completes that many simultaneous 4K requests under an SLO.

5.2 Quality evidence

Figure 3 reports the corrected GSM8K analysis. At 2B, state-only changes accuracy by -1.000 percentage point (95% CI $[-2.044, +0.044]$, nominal $p = 0.0582$). The equal-item sensitivity estimate is -0.778 points ($[-1.958, +0.410]$), with 29 gain items and 37 loss items. Int4 KV changes accuracy by -2.722 points ($[-5.275, -0.169]$, nominal $p = 0.0394$). Joint precision changes it by -1.556 points ($[-4.447, +1.336]$, nominal $p = 0.2499$). The joint-minus-KV contrast is $+1.167$ points ($[-0.987, +3.321]$). At 9B, state-only changes accuracy by $+0.333$ points ($[-0.138, +0.804]$, nominal $p = 0.1414$).

These are candidate-specific guardrail intervals, not a multiplicity-adjusted discovery family. Only the 2B int4-KV nominal interval excludes zero. We do not describe state-only or joint precision as quality preserving, beneficial, or non-inferior.

On C4, PPL changes by -0.0029, with a 95% CI from -0.0129 to $+0.0072$. On PG19, it changes by $+0.0065$, with a CI from -0.0447 to $+0.0578$. Both intervals include zero and do not establish equivalence. The harness also rounds state at chunk boundaries, so it is supporting evidence only. In five no-think RULER cells, both temporal runs observe identical fp32 and bf16 outputs across three seeds. The resulting zero-width descriptive intervals are observed equality, not population certainty (Figure 4).

An exploratory 18-layer sensitivity screen produces two raw C4 intervals with $p < 0.05$, but neither survives Bonferroni or Benjamini–Hochberg correction. This result does not show that whole-state precision is optimal. It only provides no corrected evidence for a layer exception in this screen. The Supplementary Material reports the panel and the chunk-size ablation.

5.3 Corrected selector audit

Table 3 replays the four frozen requests after replacing seed-iid quality bounds with the clustered intervals above. The strict request requires 250 allocator-equivalent slots and a -1.0 point GSM8K floor. Full precision is its only feasible candidate and has a 29.483 req/s SLO-goodput lower bound.

The medium request has no feasible candidate. Full precision provides only 269.9 slots against a requirement of 300. State-only provides 311.6 slots, but its quality lower bound is -2.044 points against a -2.0 point floor. KV-only and joint also violate quality. For the high request, joint provides enough slots but misses the -4.0 point quality floor. The negative control requires 900 slots, which no candidate provides. We did not relax any threshold after seeing these outcomes.

This audit invalidates the earlier three-mapping narrative. It demonstrates fail-closed constraint evaluation and complete rejection traces. It does not demonstrate an optimal online policy, held-out regret, or a serving advantage.

5.4 Serving boundary

Figure 5 shows the formal run and temporal rerun for Random60 and ShareGPT300. None of the 60 cellwise comparisons survives Benjamini– Hochberg false-discovery-rate control. Random60 overload cells have consistent direc-

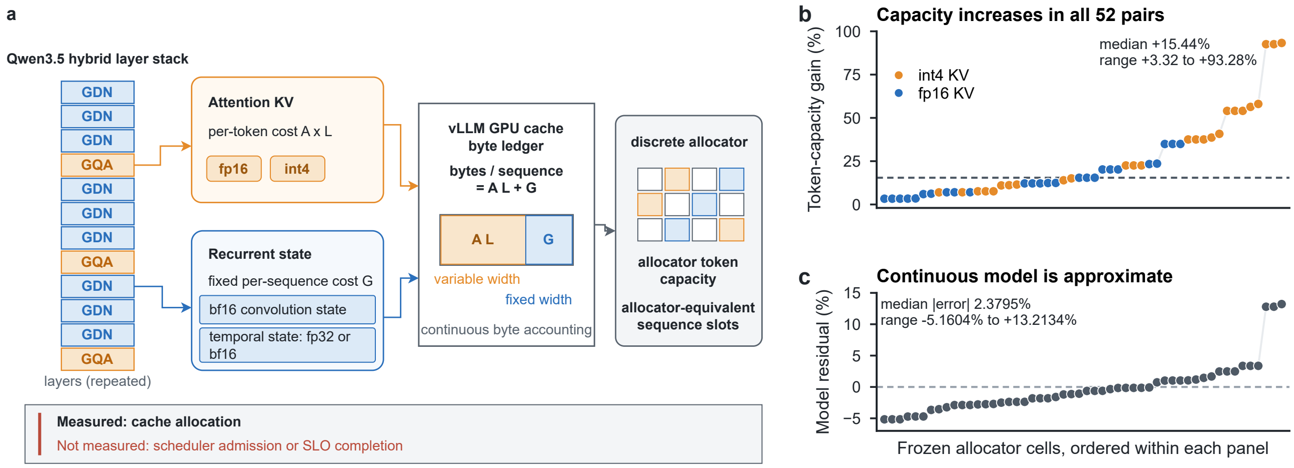

Figure 1. **From hybrid memory to measured allocator capacity.** (a) GQA layers contribute per-token attention KV, while GDN layers contribute fixed recurrent state; both enter one GPU cache byte ledger before discrete allocation. The diagram explicitly separates measured cache allocation from unmeasured scheduler admission and SLO completion. (b) All 52 paired cells increase allocator token capacity under bf16 state. (c) The continuous model has mixed-sign approximation residuals and is neither a lower nor upper bound.

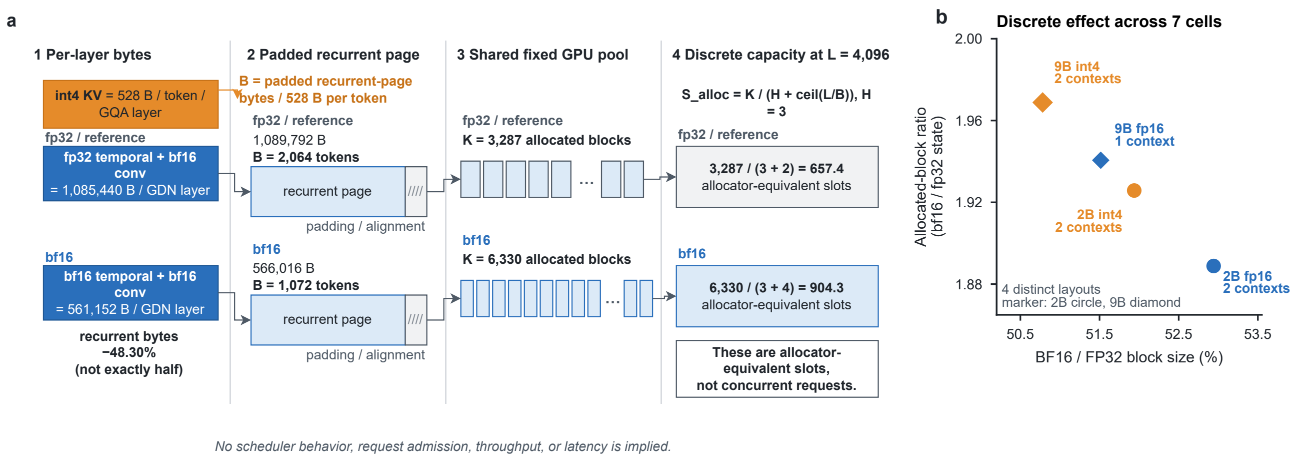

Figure 2. **Byte-to-page-to-block allocation.** (a) Verified 2B/int4/4K worked example. The recurrent-page bytes and 528-byte int4 per-token cost determine attention block size; padding and integer block counts then determine allocator-equivalent slots. The 657.4 and 904.3 values are not concurrent requests. (b) Four distinct discrete layouts cover the seven fixed-utilization cells; marker area records whether one or two context lengths share the layout.

tions across runs, while ShareGPT directions do not. We therefore do not claim a workload-general goodput effect.

The controlled mechanism matrix completes 18 cells in each run. Throughput is within 10% in 18/18 temporal comparisons, TTFT in 10/18, and TPOT in 17/18. None of nine contrast tests survives Holm correction. The broader four-configuration matrix completes 144/144 cells per run, yet fails its primary stability gate: 183/720 continuous-goodput comparisons exceed the 10% tolerance. Binary SLO labels are more stable, but they do not override the failed primary endpoint.

Table 3. Frozen selector audit after dependence-aware GSM8K reanalysis. Quality values are percentage-point differences from full precision. Slots are allocator-equivalent sequence slots, not demonstrated scheduler concurrency.

Budget	Candidate	Required slots	Quality floor (pp)	Quality 95% CI (pp)	Goodput LCB (req/s)	Outcome
strict	full	250	-1.0	[+0.00, +0.00]	29.483	**selected**
	kv_only	250	-1.0	[-5.28, -0.17]	29.358	reject: quality
	state_only	250	-1.0	[-2.04, +0.04]	29.487	reject: quality
	joint	250	-1.0	[-4.45, +1.34]	29.357	reject: quality
medium	full	300	-2.0	[+0.00, +0.00]	29.483	reject: slots
	kv_only	300	-2.0	[-5.28, -0.17]	29.358	reject: quality
	state_only	300	-2.0	[-2.04, +0.04]	29.487	reject: quality
	joint	300	-2.0	[-4.45, +1.34]	29.357	reject: quality
high	full	700	-4.0	[+0.00, +0.00]	7.948	reject: slots, TTFT
	kv_only	700	-4.0	[-5.28, -0.17]	32.137	reject: slots, TTFT, quality
	state_only	700	-4.0	[-2.04, +0.04]	15.356	reject: slots, TTFT
	joint	700	-4.0	[-4.45, +1.34]	37.624	reject: quality
negative	full	900	-4.0	[+0.00, +0.00]	7.948	reject: slots, TTFT
	kv_only	900	-4.0	[-5.28, -0.17]	32.137	reject: slots, TTFT, quality
	state_only	900	-4.0	[-2.04, +0.04]	15.356	reject: slots, TTFT
	joint	900	-4.0	[-4.45, +1.34]	37.624	reject: slots, quality

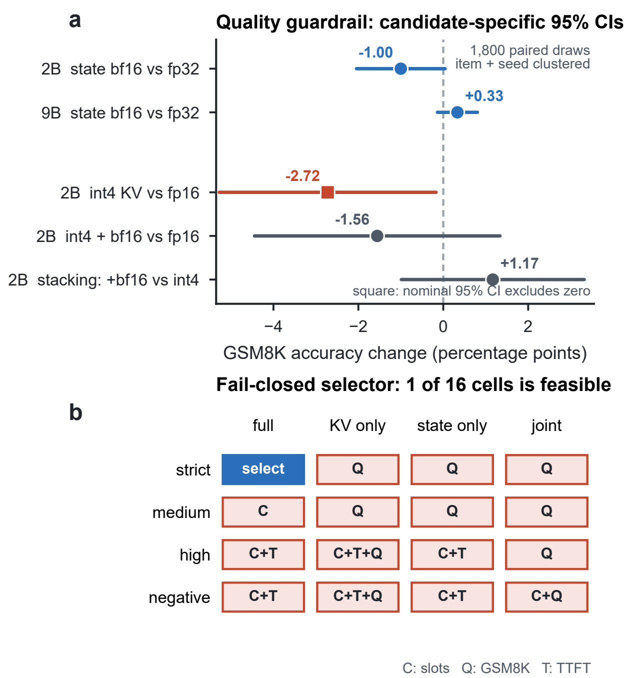

Figure 3. **Quality evidence and selector consequence.** (a) The primary GSM8K analysis uses 1,800 paired seed-item draws, 1,017 item clusters, nine dataset-seed clusters, two-way CR1 covariance, and a Student-t reference with 8 df. Squares mark a nominal 95% interval excluding zero; color is redundant. No multiplicity adjustment is applied across the displayed contrasts. (b) Complete frozen selector audit. C, Q, and T denote rejection by allocator slots, GSM8K quality, and TTFT, respectively. Only strict/full is feasible and selected.

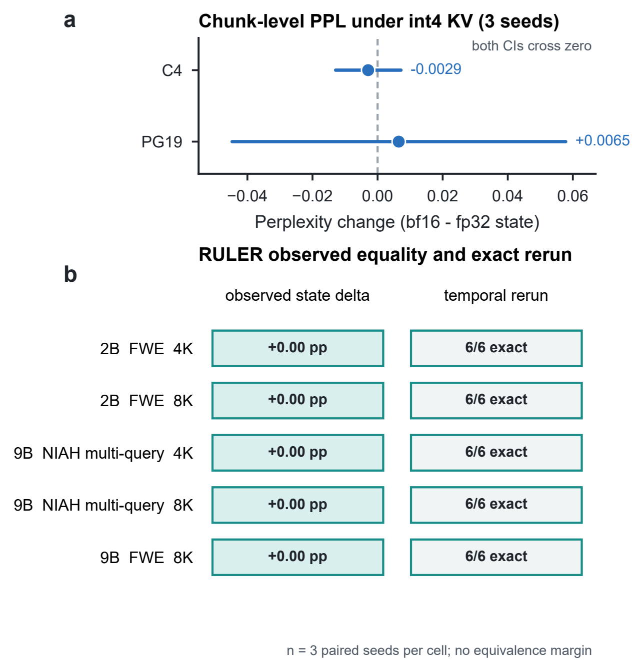

Figure 4. **PPL and RULER boundaries.** (a) Chunk-level PPL contrasts for C4 and PG19. (b) Five no-think RULER cells have exact observed equality; each cell also has 6/6 exact temporal matches across three seeds and two allocations (30/30 total). Zero-width descriptive intervals and exact reruns are not an equivalence or non-inferiority test.

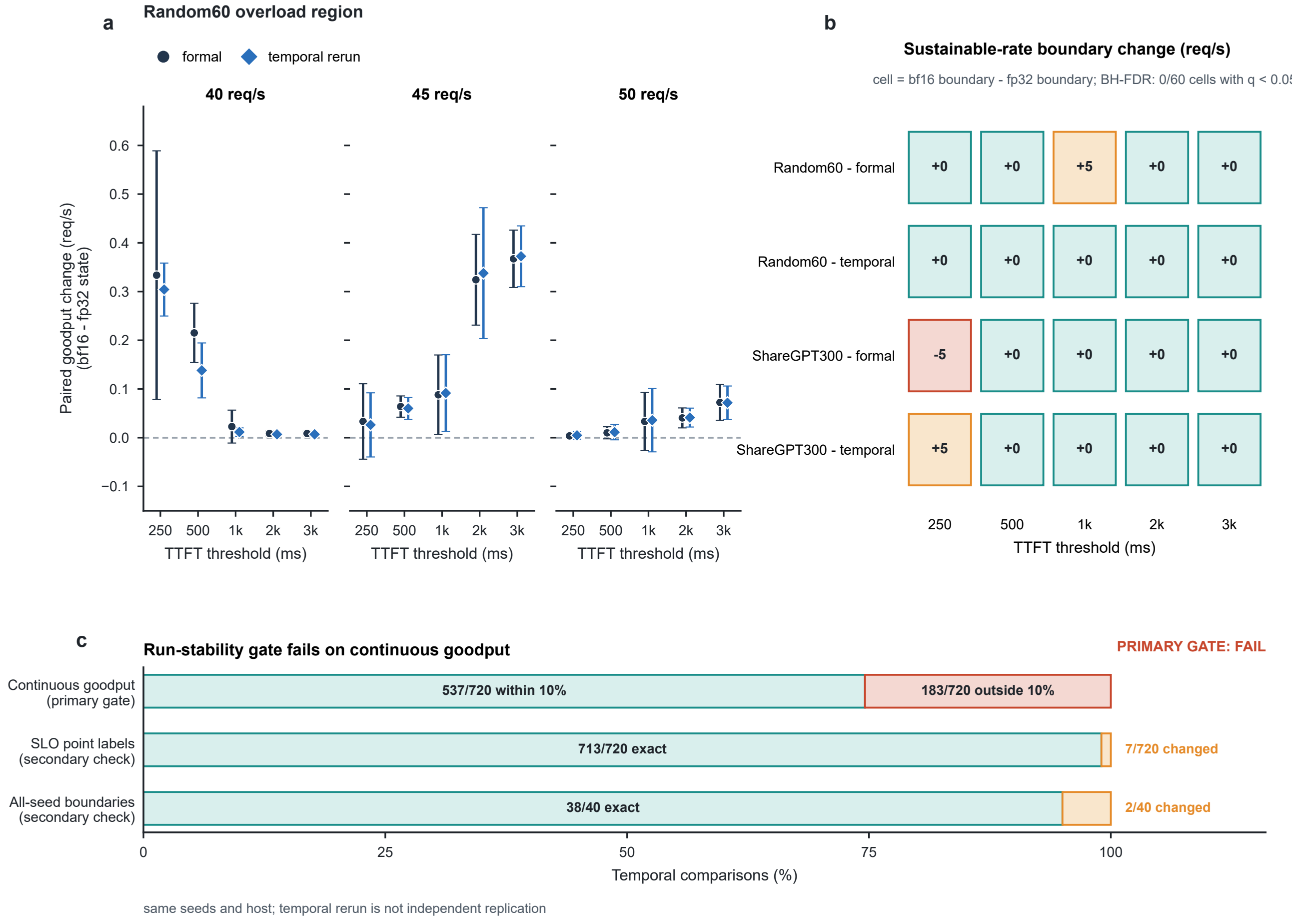

Figure 5. **Serving run-stability evidence.** (a) Random60 paired goodput change in req/s for the formal run and temporal rerun. (b) Sustainable-rate boundary changes for Random60 and ShareGPT300. The rerun uses the same host, contracts, and seeds. None of 60 cellwise comparisons survives BH-FDR. (c) The four-configuration temporal audit fails its predeclared continuous- goodput gate: 183/720 comparisons exceed 10%. Exact binary SLO labels (713/720) do not override the failed primary endpoint.

6 Discussion and Limitations

Allocator capacity is not serving concurrency. The engine's token ceiling and Eq. (3) describe cache allocation. They do not measure request admission, prefill, decode, completion, out-of-memory boundaries, or SLO success at the reported slot count. This is the central scope boundary of the paper.

The continuous model is approximate. It omits discrete attention blocks, recurrent-page padding, and layout alignment. Correct byte accounting reduces state by 48.30%, not 50%. The observed residual range has both signs, so extrapolation outside measured strata requires a fresh probe.

Quality inference remains scoped. Two-way clustering addresses repeated items and shared dataset seeds, but repeated greedy evaluations still show inconsistent outcomes for some items. The 2B baseline has 30 such repeated items and state-only has 27. The item bootstrap targets a different equal-item estimand and gives a similar state estimate, but it does not explain this variability. The contrasts are also not adjusted as one discovery family.

The selector is an audited lookup policy. It uses cold-start, exact-stratum profiles and no interpolation. Only one of four frozen requests is feasible after correction, and it selects full precision. We do not compare against an oracle, a greedy baseline, or held-out strata. Online adaptation and switching overhead are unmeasured.

System and hardware coverage is narrow. All measurements use Qwen3.5-2B/9B, one RTX 5090, and one vLLM lineage. Tensor parallelism, prefix caching, offloading, other hybrid architectures, and other accelerators can change both terms and rounding. The PPL harness is not a per-token kernel equivalence test, and RULER has only three dataset seeds.

Artifact status is local. The current anonymous draft has no public artifact URL. Reproducibility claims refer to repository-local JSON, CSV, scripts, and SHA-256 sidecars. The figure verifier pins 16 source hashes, checks all 52 paired capacity rows, and fails on numeric drift. Public release and independent-host validation remain future work.

7 Conclusion

Hybrid LLM cache accounting must include both per-token attention KV and fixed recurrent state. In the measured Qwen3.5/vLLM matrix, bf16 temporal state raises allocator token capacity in all 52 pairs, with a 15.44% median increase. The effect is strongest at short contexts, but the continuous model retains material, mixed-sign residuals.

The dependence-aware quality audit changes the deployment conclusion. The state-only GSM8K interval includes zero but extends below the strict and medium quality floors. Consequently only the strict request remains feasible, and it selects full precision. The serving matrix provides no stable end-to-end benefit. The defensible result is therefore an allocator characterization and an auditable example of why capacity, quality dependence, and operational serving evidence must remain separate.

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